```
# This Python 3 environment comes with many helpful analytics
libraries installed
# It is defined by the kaggle/python Docker image:
https://github.com/kaggle/docker-python
# For example, here's several helpful packages to load

import numpy as np # linear algebra
import pandas as pd # data processing, CSV file I/O (e.g. pd.read_csv)

# Input data files are available in the read-only "../input/"
directory
# For example, running this (by clicking run or pressing Shift+Enter)
will list all files under the input directory

import os
for dirname, _, filenames in os.walk('/kaggle/input'):
    for filename in filenames:
        print(os.path.join(dirname, filename))

# You can write up to 20GB to the current directory (/kaggle/working/)
that gets preserved as output when you create a version using "Save &
Run All"
# You can also write temporary files to /kaggle/temp/, but they won't
be saved outside of the current session
```

```
/kaggle/input/datasets/mahmoudalimahmoud/students-data/Students -
Students.csv
```

Import Libraries

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import plotly.express as px
```

Reading Dataset

```
data = pd.read_csv("/kaggle/input/datasets/mahmoudalimahmoud/students-
data/Students - Students.csv")
df = data.copy()
df
```

```
      student_id  age gender academic_level  study_hours
self_study_hours  \
0              1   18  Other    High School         7.64
1.56
```

```
1              2   18  Other    High School         2.21
2.22
2              3   22   Male    High School         3.45
0.00
3              4   17  Other    High School         5.75
2.08
4              5   19  Other    High School         6.83
1.72
...          ...  ...    ...            ...          ...
...
4995        4996   20  Other  Undergraduate         3.93
1.42
4996        4997   24   Male    High School         4.00
0.00
4997        4998   22   Male  Undergraduate         5.63
2.52
4998        4999   17   Male  Undergraduate         3.58
2.83
4999        5000   22  Other  Undergraduate         5.60
4.14

      online_classes_hours  social_media_hours  gaming_hours
sleep_hours  \
0                     2.20                3.05          2.19
6.52
1                     2.10                1.65          2.55
5.97
2                     0.29                1.34          2.08
8.39
3                     3.01                2.27          2.20
6.31
4                     3.33                2.65          0.70
8.01
...                    ...                 ...           ...
...
4995                  0.54                2.58          2.62
9.30
4996                  3.00                4.45          2.47
6.42
4997                  2.89                3.58          1.07
6.65
4998                  1.53                5.88          1.28
7.36
4999                  2.17                3.05          2.74
6.38

      ...  exercise_minutes  caffeine_intake_mg  part_time_job  \
0     ...                81                  38              1
1     ...               111                 339              0
```

```
2     ...                 68                  266              0
3     ...                113                  480              1
4     ...                121                   24              1
...   ...                ...                  ...            ...
4995  ...                113                  496              0
4996  ...                146                  411              0
4997  ...                123                  306              0
4998  ...                 54                  304              1
4999  ...                 24                  219              0

      upcoming_deadline  internet_quality mental_health_score
focus_index  \
0                     0              Good                  10
43.05
1                     0              Good                   3
15.92
2                     1              Good                   8
27.39
3                     1              Poor                   3
22.31
4                     0              Good                   8
38.11
...                 ...               ...                 ...
...
4995                  0              Good                   9
35.55
4996                  1              Poor                   5
16.87
4997                  1              Good                   7
23.85
4998                  1              Good                   4
13.04
4999                  1              Good                   3
29.98

      burnout_level  productivity_score  exam_score
0             31.77               73.65       50.16
1             37.00               13.70        1.00
2             34.37               45.15       18.30
3             77.31               20.92        9.37
4             39.53               59.23       27.81
...             ...                 ...         ...
4995          33.24               48.83       22.51
4996          59.07               23.32        2.00
4997          47.72               41.09       27.89
4998          58.90               12.62        3.63
4999          61.01               34.72       10.76

[5000 rows x 21 columns]
```

Sample Of Data

```
print("Data Sample : \n")
df.sample()
```

```
Data Sample :


      student_id  age  gender academic_level  study_hours
self_study_hours  \
1474        1475   19  Female    High School         5.29
3.1

      online_classes_hours  social_media_hours  gaming_hours
sleep_hours  \
1474                   4.8                1.18          2.26
5.95

      ...  exercise_minutes  caffeine_intake_mg  part_time_job  \
1474  ...                16                 494              0

      upcoming_deadline  internet_quality mental_health_score
focus_index  \
1474                  0           Average                   4
33.0

      burnout_level  productivity_score  exam_score
1474           33.8               47.33       43.03

[1 rows x 21 columns]
```

EDA (Exploratry Data Analysis)

Data Information

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 5000 entries, 0 to 4999
Data columns (total 21 columns):
 #   Column                Non-Null Count  Dtype
---  ------                --------------  -----
 0   student_id            5000 non-null   int64
 1   age                   5000 non-null   int64
 2   gender                5000 non-null   object
 3   academic_level        5000 non-null   object
 4   study_hours           5000 non-null   float64
```

```
 5   self_study_hours      5000 non-null   float64
 6   online_classes_hours  5000 non-null   float64
 7   social_media_hours    5000 non-null   float64
 8   gaming_hours          5000 non-null   float64
 9   sleep_hours           5000 non-null   float64
 10  screen_time_hours     5000 non-null   float64
 11  exercise_minutes      5000 non-null   int64
 12  caffeine_intake_mg    5000 non-null   int64
 13  part_time_job         5000 non-null   int64
 14  upcoming_deadline     5000 non-null   int64
 15  internet_quality      5000 non-null   object
 16  mental_health_score   5000 non-null   int64
 17  focus_index           5000 non-null   float64
 18  burnout_level         5000 non-null   float64
 19  productivity_score    5000 non-null   float64
 20  exam_score            5000 non-null   float64
dtypes: float64(11), int64(7), object(3)
memory usage: 820.4+ KB
```

Data Describtion

```
df.describe()
```

```
        student_id          age  study_hours  self_study_hours  \
count  5000.000000  5000.000000  5000.000000       5000.000000
mean   2500.500000    20.520400     4.539594          2.478734
std    1443.520003     2.870406     1.821665          1.177990
min       1.000000    16.000000     0.000000          0.000000
25%    1250.750000    18.000000     3.250000          1.660000
50%    2500.500000    20.000000     4.530000          2.480000
75%    3750.250000    23.000000     5.760000          3.290000
max    5000.000000    25.000000    11.840000          7.410000

       online_classes_hours  social_media_hours  gaming_hours
sleep_hours  \
count           5000.000000         5000.000000   5000.000000
5000.000000
mean               2.011984            2.998086      1.564514
7.016492
std                0.983906            1.467949      1.110807
1.163692
min                0.000000            0.000000      0.000000
4.000000
25%                1.320000            1.990000      0.670000
6.237500
50%                2.010000            2.980000      1.490000
7.010000
75%                2.690000            4.030000      2.340000
```

```
7.810000
max                6.000000            8.280000      5.640000
10.000000

       screen_time_hours  exercise_minutes  caffeine_intake_mg
part_time_job  \
count        5000.000000       5000.000000         5000.000000
5000.000000
mean            6.979588         74.535600          251.450400
0.498200
std             2.486214         42.932293          143.842712
0.500047
min             1.000000          0.000000            0.000000
0.000000
25%             5.280000         37.000000          129.000000
0.000000
50%             6.950000         75.000000          252.000000
0.000000
75%             8.710000        112.000000          376.000000
1.000000
max            15.300000        149.000000          499.000000
1.000000

       upcoming_deadline  mental_health_score  focus_index
burnout_level  \
count        5000.000000          5000.000000  5000.000000
5000.000000
mean            0.501400             5.507400    29.431616
45.615324
std             0.500048             2.869145     9.962902
14.246591
min             0.000000             1.000000     1.000000
1.000000
25%             0.000000             3.000000    22.567500
35.727500
50%             1.000000             5.000000    29.430000
45.690000
75%             1.000000             8.000000    36.242500
55.352500
max             1.000000            10.000000    63.480000
97.580000

       productivity_score   exam_score
count         5000.000000  5000.000000
mean            37.267716    18.803752
std             16.849397    12.130840
min              1.000000     1.000000
25%             25.290000     9.337500
50%             36.860000    18.010000
```

```
75%             49.142500    27.400000
max             98.020000    64.090000
```

```
print("describtion for object data : \n")
df.describe(include = "O")
```

```
describtion for object data :


       gender academic_level internet_quality
count    5000           5000             5000
unique      3              3                3
top      Male   Postgraduate             Good
freq     1719           1687             1722
```

Data Shape

```
df.shape
```

```
(5000, 21)
```

```
print(f"number of rows in datast = {df.shape[0]}")
print(f"number of columns in datast = {df.shape[1]}")
```

```
number of rows in datast = 5000
number of columns in datast = 21
```

```
print("first 5 rows in dataset")
df.head(5)
```

```
first 5 rows in dataset

   student_id  age gender academic_level  study_hours
self_study_hours  \
0           1   18  Other    High School         7.64
1.56
1           2   18  Other    High School         2.21
2.22
2           3   22   Male    High School         3.45
0.00
3           4   17  Other    High School         5.75
2.08
4           5   19  Other    High School         6.83
1.72

   online_classes_hours  social_media_hours  gaming_hours  sleep_hours
...  \
0                  2.20                3.05          2.19         6.52
...
1                  2.10                1.65          2.55         5.97
```

```
...
2                  0.29                 1.34          2.08          8.39
...
3                  3.01                 2.27          2.20          6.31
...
4                  3.33                 2.65          0.70          8.01
...

   exercise_minutes  caffeine_intake_mg  part_time_job
upcoming_deadline  \
0                 81                  38              1
0
1                111                 339              0
0
2                 68                 266              0
1
3                113                 480              1
1
4                121                  24              1
0

   internet_quality mental_health_score  focus_index  burnout_level  \
0              Good                  10        43.05          31.77
1              Good                   3        15.92          37.00
2              Good                   8        27.39          34.37
3              Poor                   3        22.31          77.31
4              Good                   8        38.11          39.53

   productivity_score  exam_score
0               73.65       50.16
1               13.70        1.00
2               45.15       18.30
3               20.92        9.37
4               59.23       27.81

[5 rows x 21 columns]
```

Data Cleaning & Preprocessing

```
print('Missing Values in Each Column: \n')
df.isna().sum()
```

```
Missing Values in Each Column:


student_id               0
age                      0
gender                   0
```

```
academic_level           0
study_hours              0
self_study_hours         0
online_classes_hours     0
social_media_hours       0
gaming_hours             0
sleep_hours              0
screen_time_hours        0
exercise_minutes         0
caffeine_intake_mg       0
part_time_job            0
upcoming_deadline        0
internet_quality         0
mental_health_score      0
focus_index              0
burnout_level            0
productivity_score       0
exam_score               0
dtype: int64
```

```
df.duplicated().sum()
```

```
np.int64(0)
```

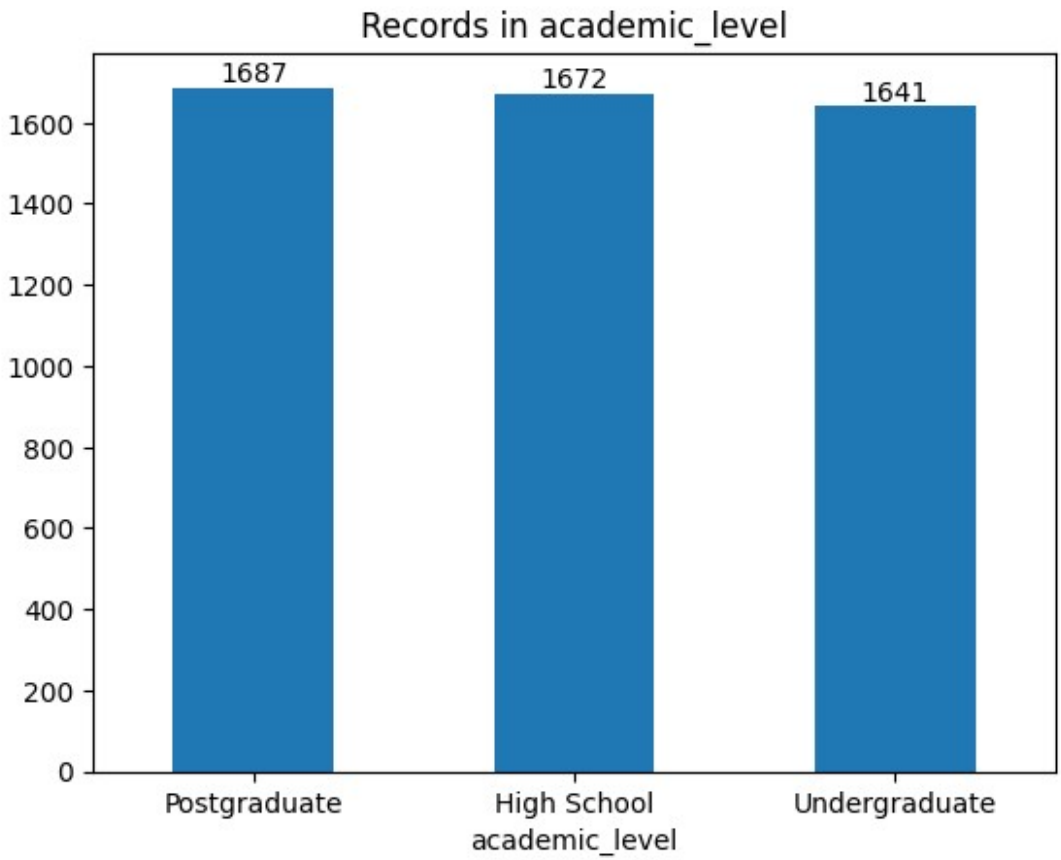
```
print("Records in academic_level")

df["academic_level"].value_counts()
```

```
Records in academic_level

academic_level
Postgraduate     1687
High School      1672
Undergraduate    1641
Name: count, dtype: int64
```

```
a = df["academic_level"].value_counts().plot(kind="bar")
for container in a.containers:
    a.bar_label(container)
plt.title("Records in academic_level")
plt.xticks(rotation = 360)
plt.show()
```



Possible Analysis Questions

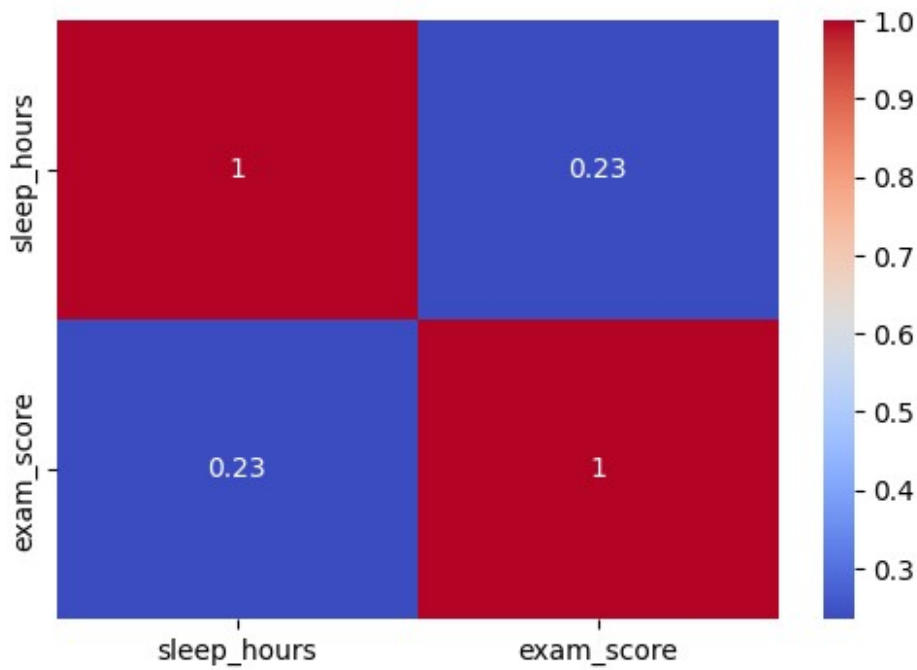
1• How does sleep affect exam performance?

1. weak positive correlation

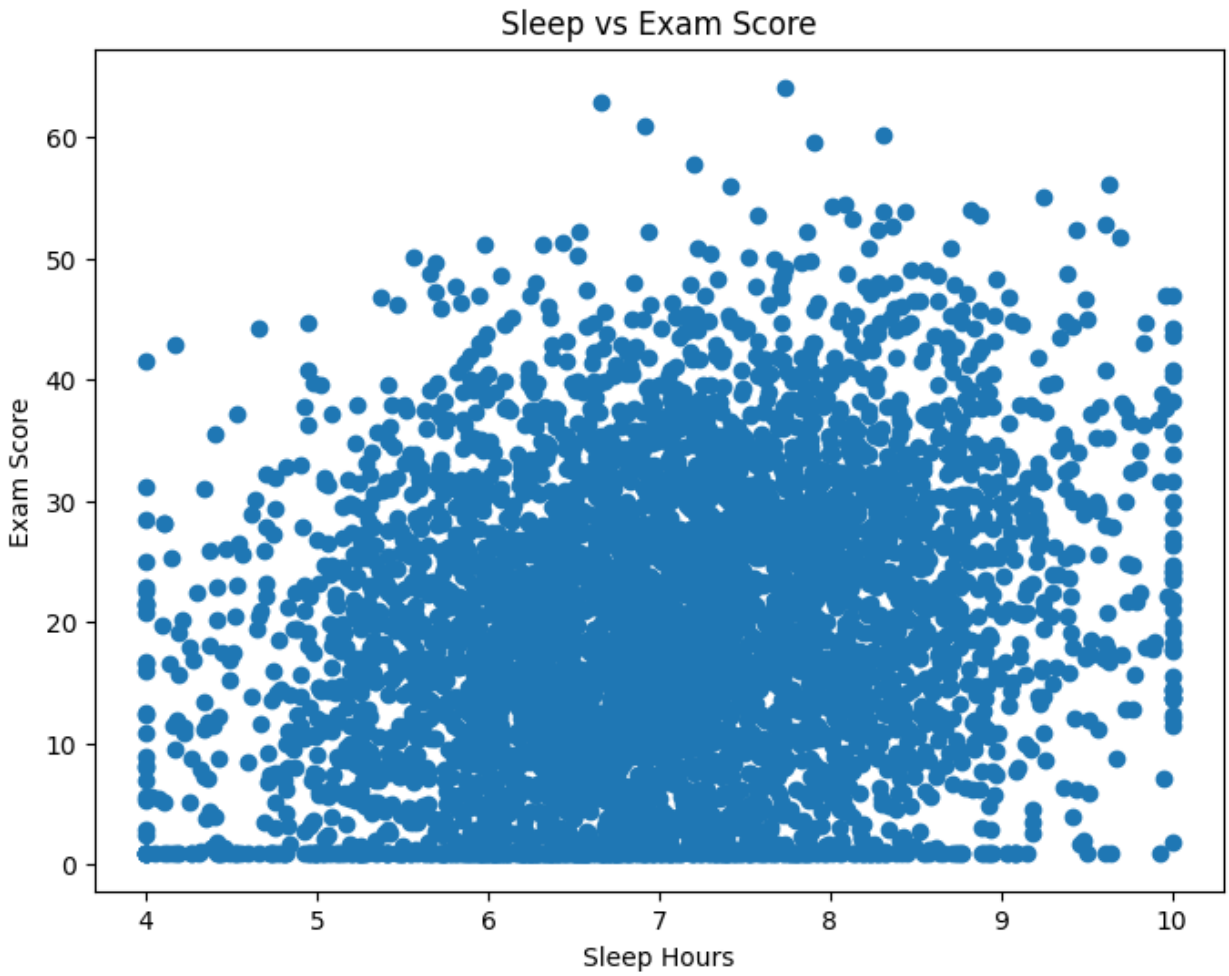
```
sleep_hours_score = df[["sleep_hours","exam_score"]].corr()
sleep_hours_score
```

```
              sleep_hours  exam_score
sleep_hours       1.00000     0.23471
exam_score        0.23471     1.00000
```

```
plt.figure(figsize=(6,4))
sns.heatmap(sleep_hours_score,annot=True,cmap="coolwarm")
plt.show()
```




```
plt.figure(figsize=(8, 6))
plt.scatter(df["sleep_hours"], df["exam_score"])
plt.xlabel("Sleep Hours")
plt.ylabel("Exam Score")
plt.title("Sleep vs Exam Score")
plt.show()
```



or

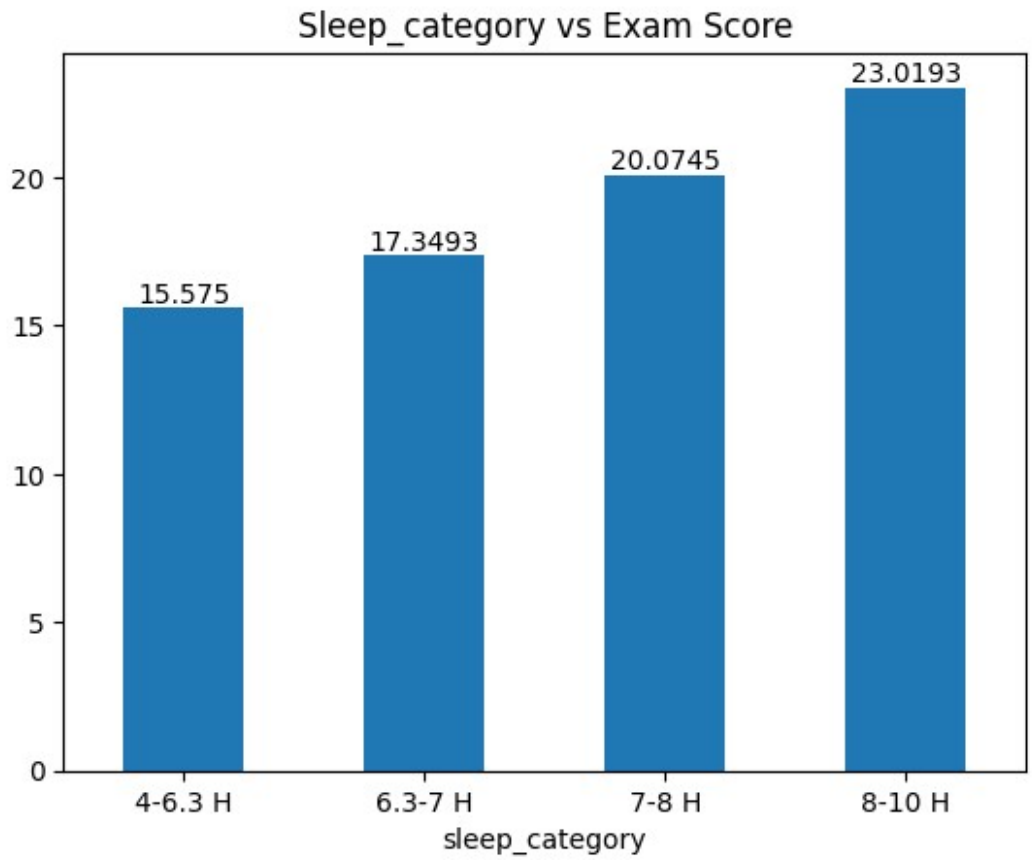
```
bins = [4,6.3,7.01,8,10]
labels = ["4-6.3 H","6.3-7 H","7-8 H","8-10 H"]
df["sleep_category"] =
pd.cut(df["sleep_hours"],bins=bins,labels=labels)

df.groupby("sleep_category",observed=True)["exam_score"].mean()

sleep_category
4-6.3 H    15.575034
6.3-7 H    17.349290
7-8 H      20.074491
8-10 H     23.019302
Name: exam_score, dtype: float64

a = df.groupby("sleep_category",observed=True)
["exam_score"].mean().plot(kind="bar")
plt.xticks(rotation = 360)
```

```
plt.title("Sleep_category vs Exam Score")
for container in a.containers:
    a.bar_label(container)
plt.show()
```



2• Does social media usage reduce productivity?

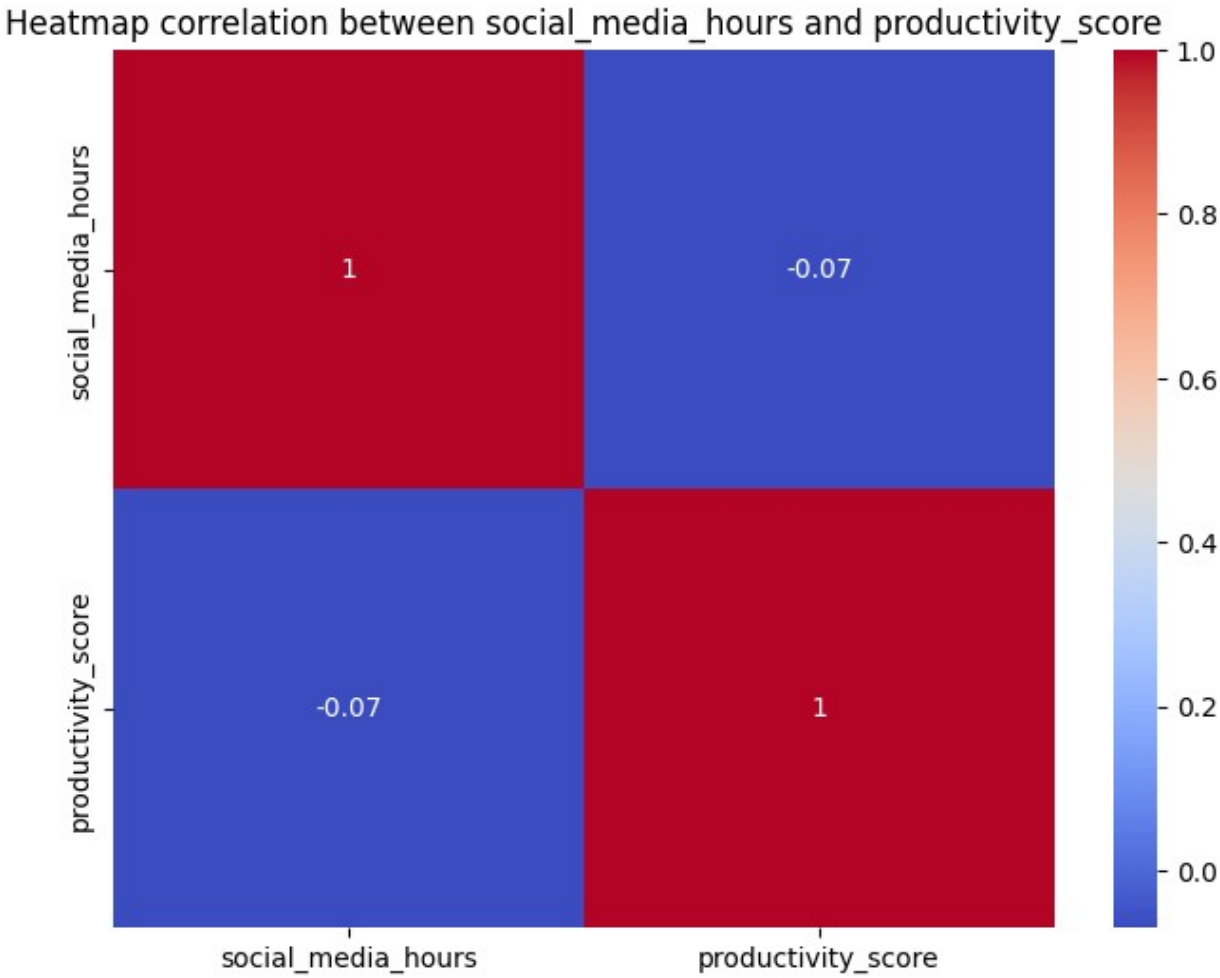
No

```
social_media_score =
df[["social_media_hours","productivity_score"]].corr()
social_media_score
```

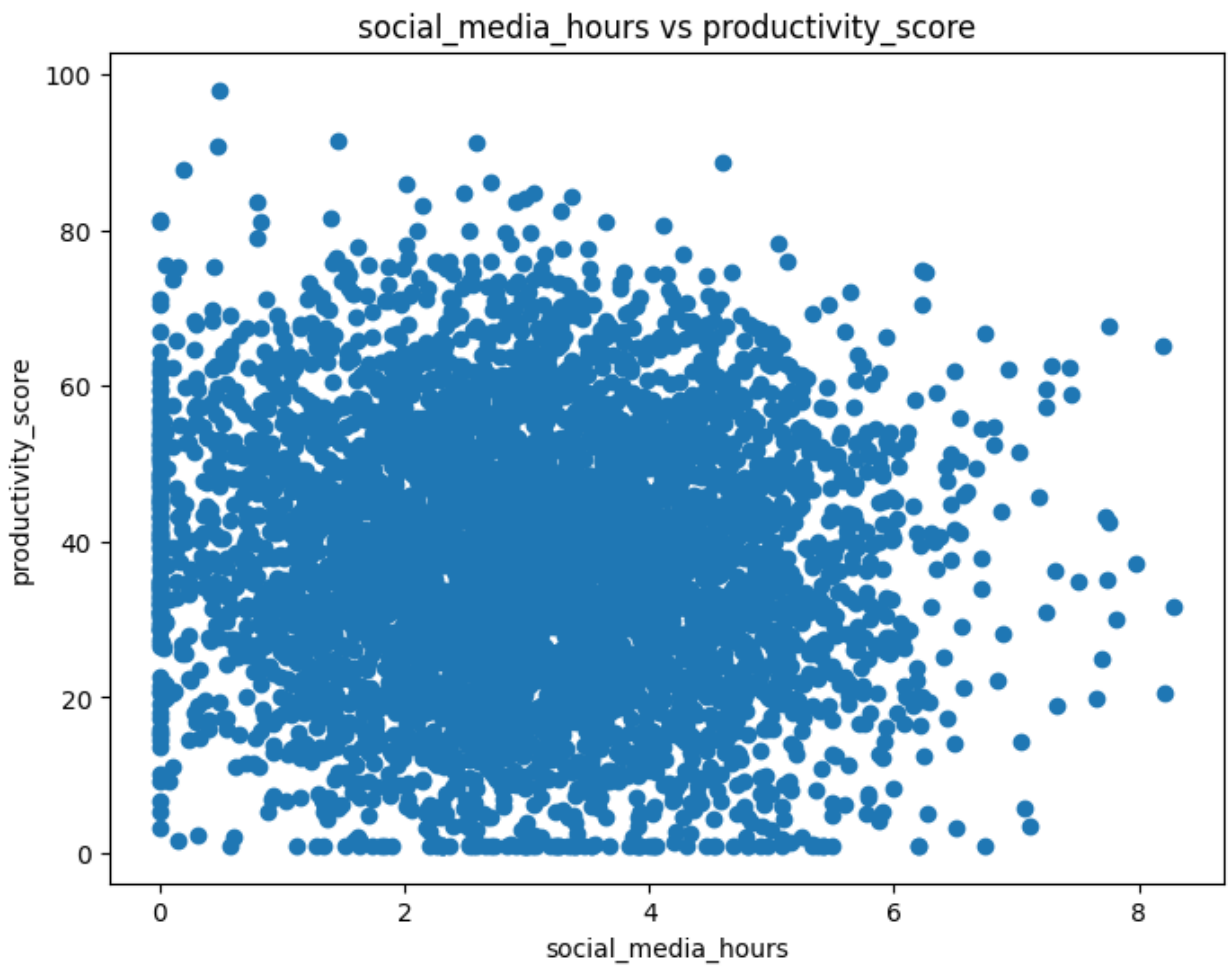
```
                     social_media_hours  productivity_score
social_media_hours            1.000000           -0.070052
productivity_score           -0.070052            1.000000
```

```
plt.figure(figsize=(8,6))
sns.heatmap(social_media_score,annot=True,cmap="coolwarm")
```

```
plt.title("Heatmap correlation between social_media_hours and
productivity_score")
plt.show()
```




```
plt.figure(figsize=(8, 6))
plt.scatter(df["social_media_hours"], df["productivity_score"])
plt.xlabel("social_media_hours")
plt.ylabel("productivity_score")
plt.title("social_media_hours vs productivity_score")
plt.show()
```



or

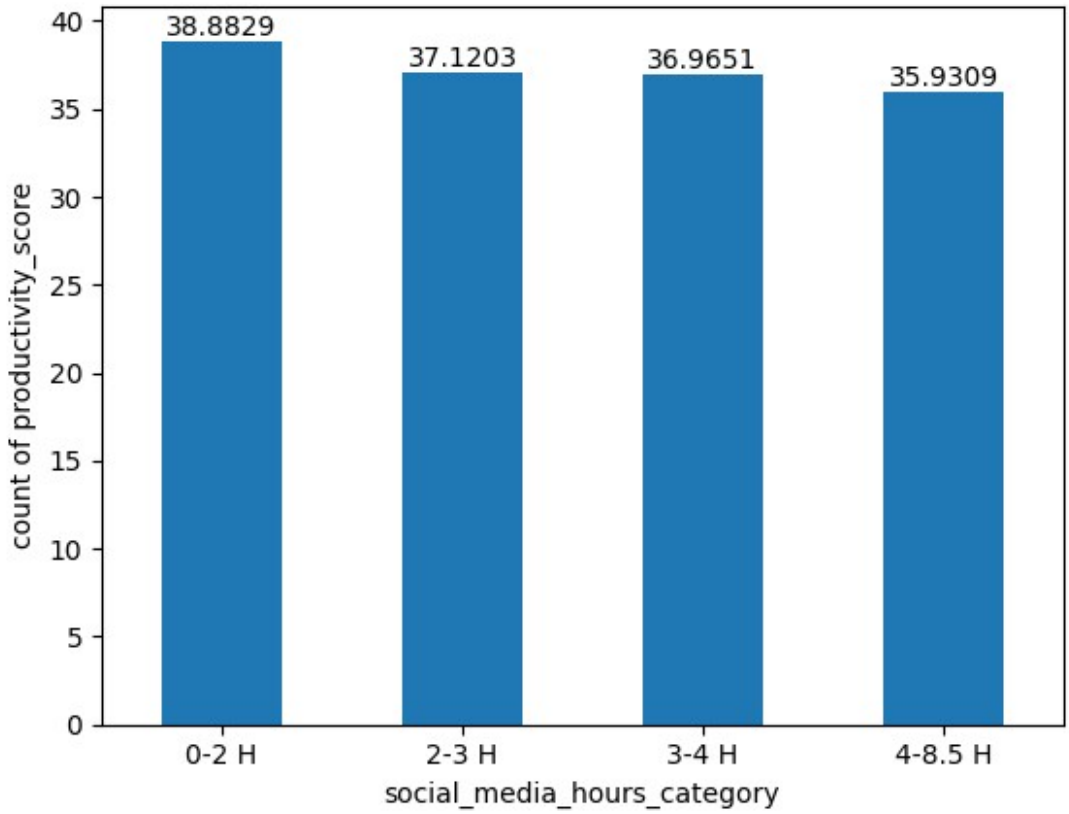
```
bins = [0,2,3,4.03,8.3]
labels = ["0-2 H","2-3 H","3-4 H","4-8.5 H"]
df["social_media_hours_category"] =
pd.cut(df["social_media_hours"],bins=bins,labels=labels)

df.groupby("social_media_hours_category",observed=True)
["productivity_score"].mean()

social_media_hours_category
0-2 H      38.882893
2-3 H      37.120268
3-4 H      36.965065
4-8.5 H    35.930911
Name: productivity_score, dtype: float64

b = df.groupby("social_media_hours_category",observed=True)
["productivity_score"].mean().plot(kind="bar")
```

```
plt.xticks(rotation = 360)
plt.title=("social_media_hours_category vs productivity_score")
plt.ylabel("count of productivity_score")
for container in b.containers:
    b.bar_label(container)
plt.show()
```



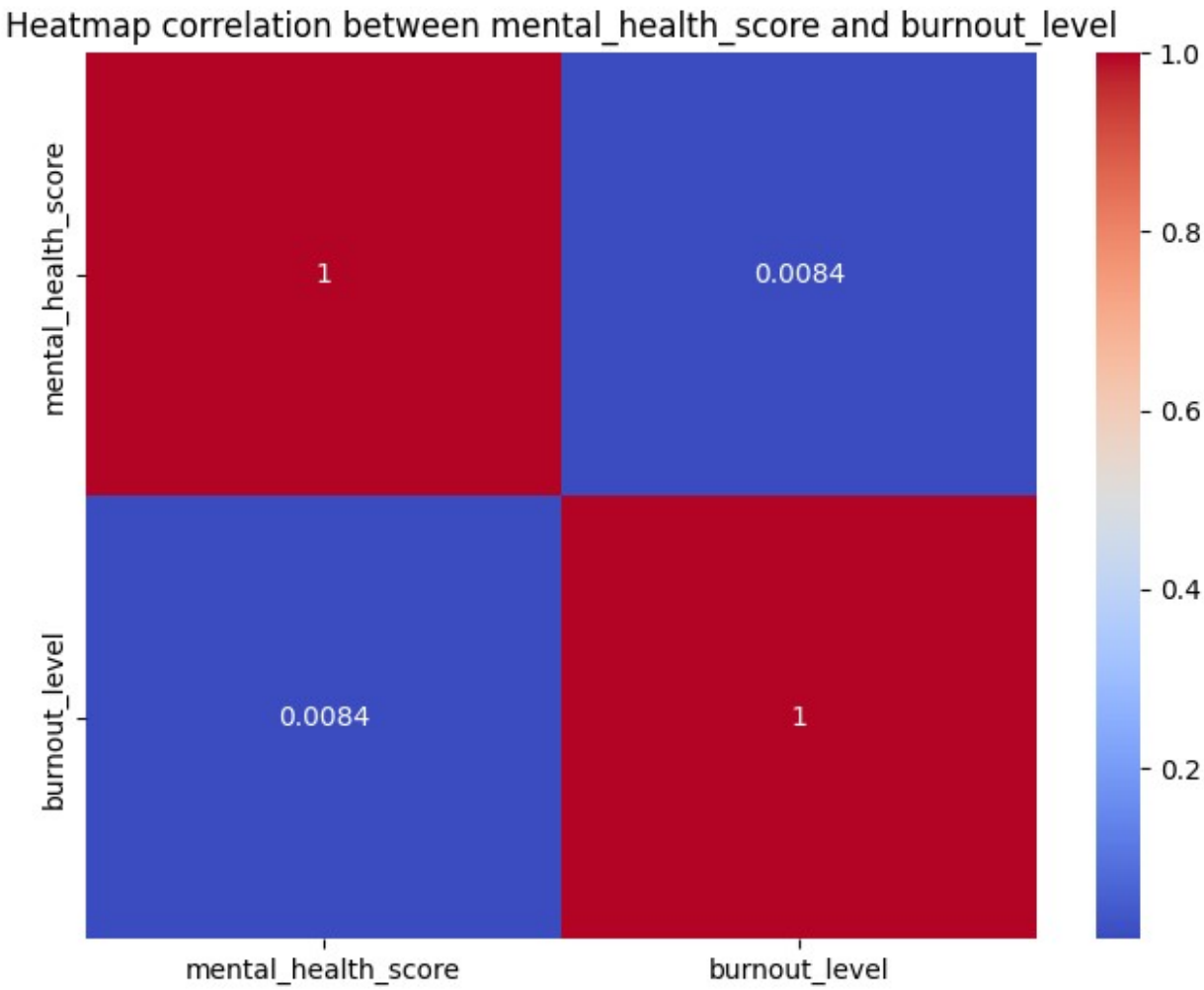
3. • Relationship between burnout and mental health.

No correlation

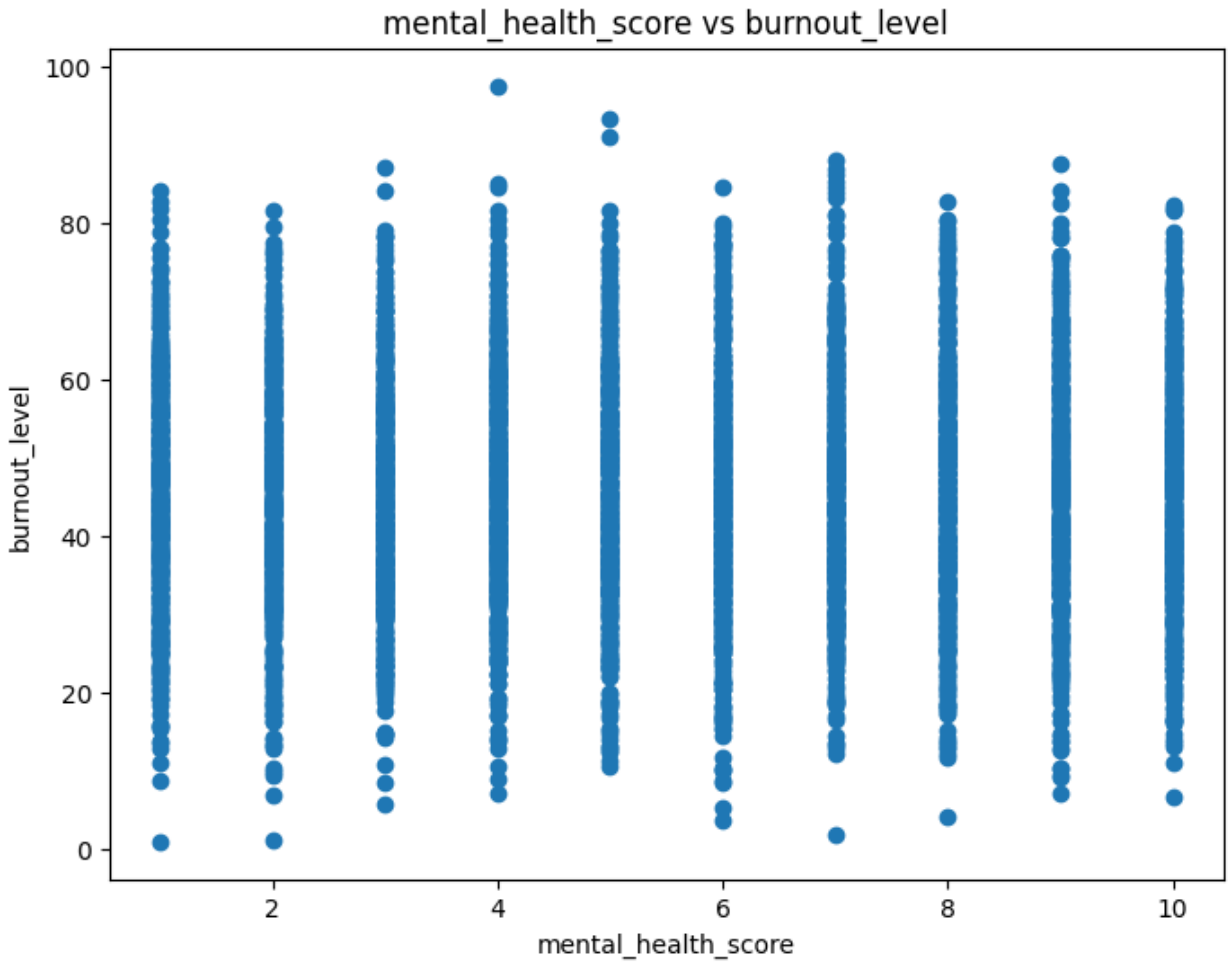
```
burnout_health_score=df[["mental_health_score","burnout_level"]].corr(
)
burnout_health_score
```

```
                      mental_health_score  burnout_level
mental_health_score             1.000000       0.008399
burnout_level                   0.008399       1.000000
```

```
plt.figure(figsize=(8,6))
sns.heatmap(burnout_health_score,annot=True,cmap="coolwarm")
plt.title("Heatmap correlation between mental_health_score and
burnout_level")
plt.show()
```




```
plt.figure(figsize=(8, 6))
plt.scatter(df["mental_health_score"], df["burnout_level"])
plt.xlabel("mental_health_score")
plt.ylabel("burnout_level")
plt.title("mental_health_score vs burnout_level")
plt.show()
```



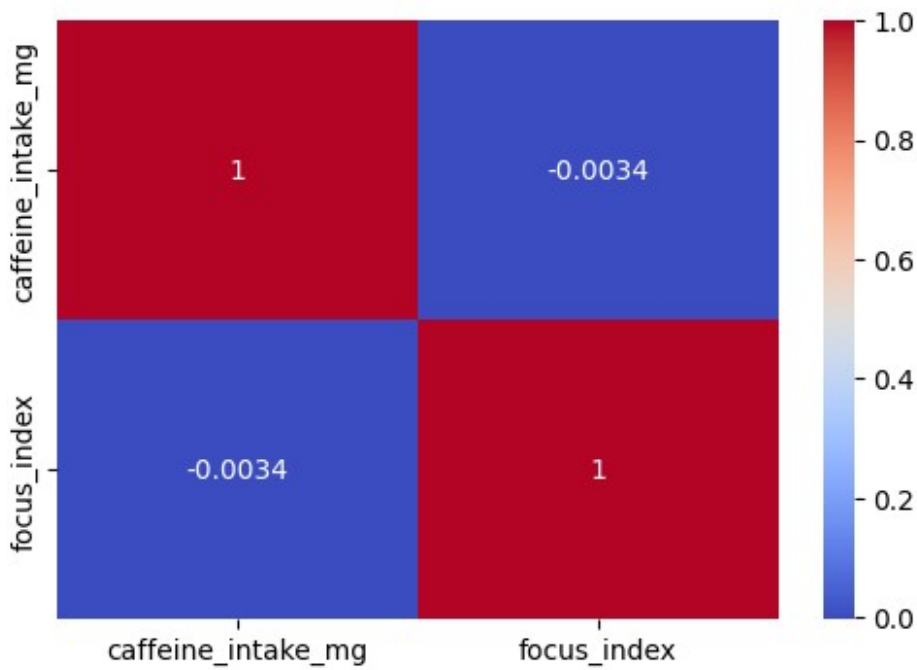
4• Does caffeine improve focus?

NO

```
caffine_focus = df[["caffeine_intake_mg","focus_index"]].corr()
caffine_focus
```

```
                     caffeine_intake_mg  focus_index
caffeine_intake_mg             1.000000    -0.003389
focus_index                   -0.003389     1.000000
```

```
plt.figure(figsize=(6,4))
sns.heatmap(caffine_focus,annot=True,cmap="coolwarm")
plt.title=("heatmap correlation caffeine_intake_mg vs focus_index")
plt.show()
```



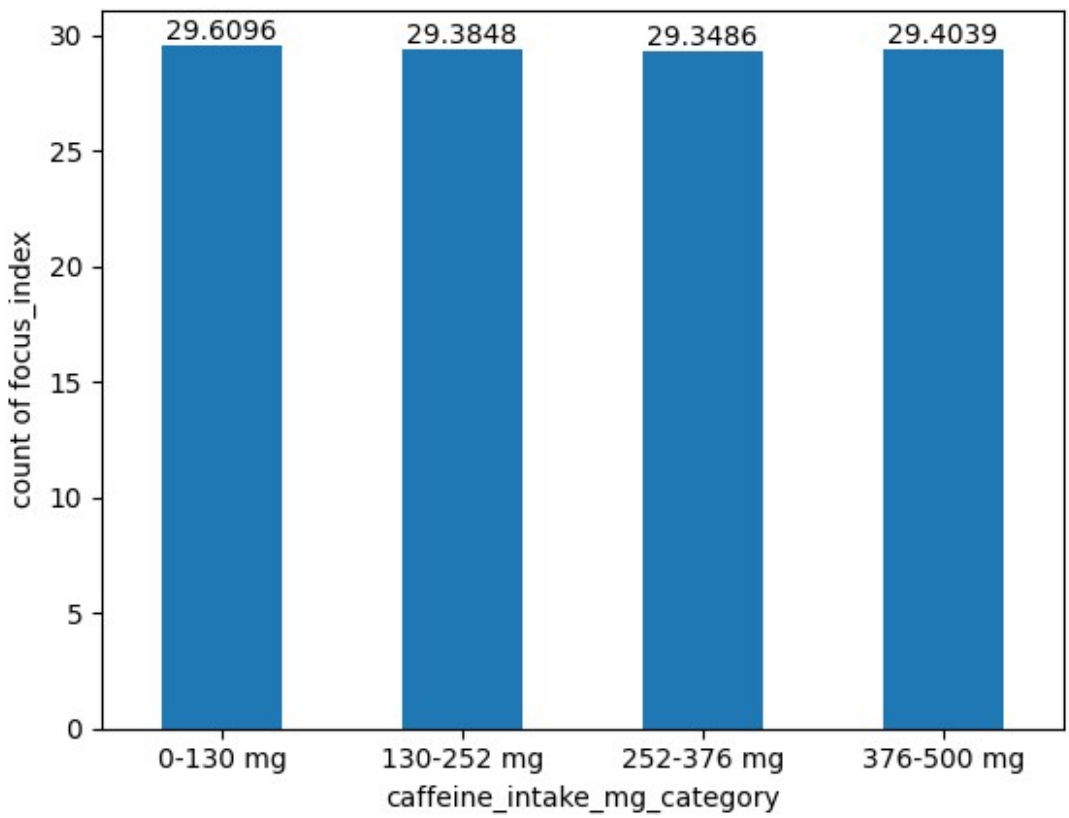
or

```
bins = [0,130,252,376,500]
labels = ["0-130 mg","130-252 mg","252-376 mg","376-500 mg"]
df["caffeine_intake_mg_category"] =
pd.cut(df["caffeine_intake_mg"],bins=bins,labels=labels)

df.groupby("caffeine_intake_mg_category",observed=True)
["focus_index"].mean()

caffeine_intake_mg_category
0-130 mg      29.609566
130-252 mg    29.384826
252-376 mg    29.348582
376-500 mg    29.403896
Name: focus_index, dtype: float64

c = df.groupby("caffeine_intake_mg_category",observed=True)
["focus_index"].mean().plot(kind="bar")
plt.xticks(rotation = 360)
plt.title=("caffeine_intake_mg_category vs focus_index")
plt.ylabel("count of focus_index")
for container in c.containers:
    c.bar_label(container)
plt.show()
```



5• Impact of part-time jobs on academic performance.

```
part_time_score = df.groupby('part_time_job')
['exam_score'].mean().reset_index()
part_time_score
```

```
   part_time_job  exam_score
0              0   20.614328
1              1   16.980092
```

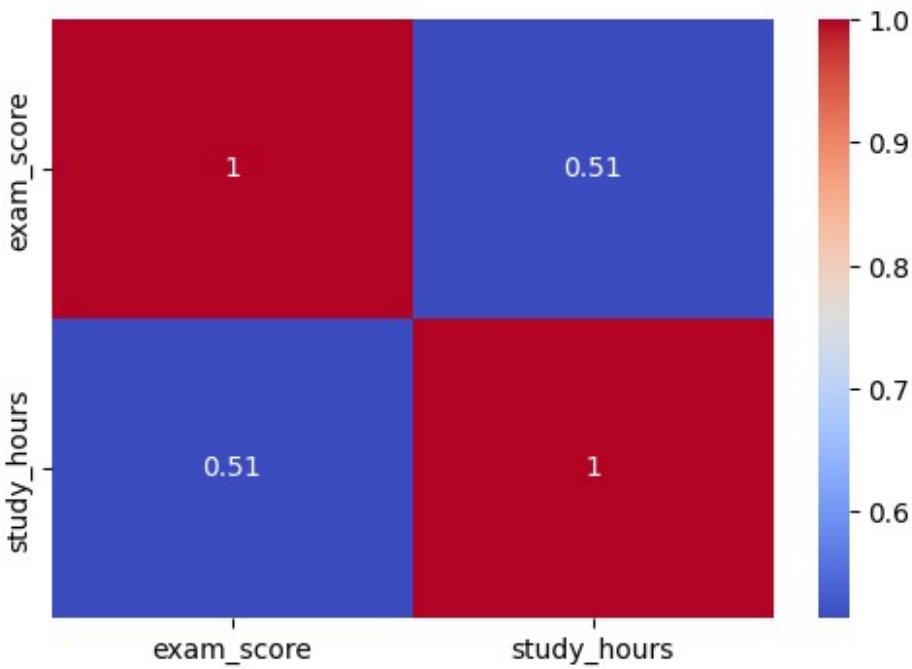
```
fig_part_time_score =
px.bar(part_time_score,x="part_time_job",y="exam_score",text_auto=True
,title="Average of exam_score for part_time_job")
fig_part_time_score.show()
```

6• Effect of study hours on exam scores.

```
study_hours_score = df[["exam_score","study_hours"]].corr()
study_hours_score
```

```
              exam_score  study_hours
exam_score     1.000000     0.513434
study_hours    0.513434     1.000000
```

```
plt.figure(figsize=(6,4))
sns.heatmap(study_hours_score,annot=True,cmap="coolwarm")
plt.show()
```

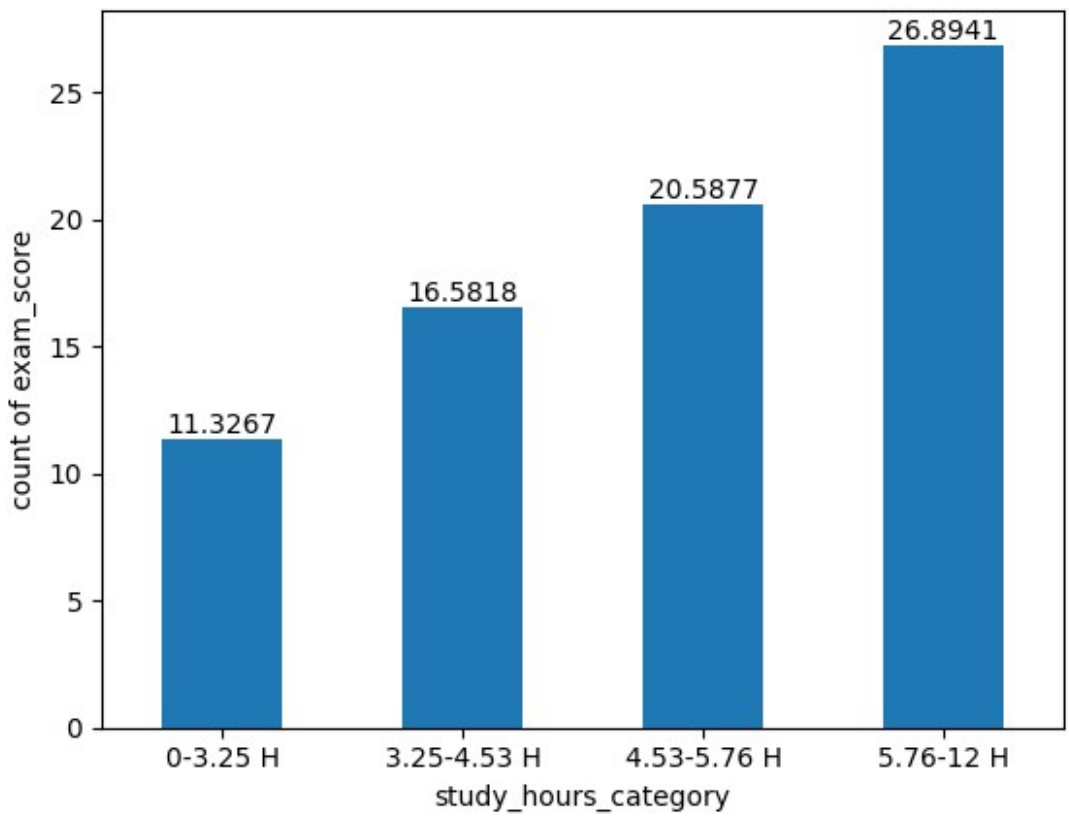


or

```
bins = [0,3.25,4.53,5.76,11.84]
labels = ["0-3.25 H","3.25-4.53 H","4.53-5.76 H","5.76-12 H"]
df["study_hours_category"] =
pd.cut(df["study_hours"],bins=bins,labels=labels)
```

```
df.groupby("study_hours_category", observed=True)["exam_score"].mean()
```

```
study_hours_category
0-3.25 H       11.326705
3.25-4.53 H    16.581809
4.53-5.76 H    20.587712
5.76-12 H      26.894147
Name: exam_score, dtype: float64
```

```
xyz = df.groupby("study_hours_category", observed=True)
["exam_score"].mean().plot(kind="bar")
plt.xticks(rotation = 360)
```

```
plt.title=("study_hours_category vs exam_score")
plt.ylabel("count of exam_score")
for container in xyz.containers:
    xyz.bar_label(container)
plt.show()
```